ALAGAPPA UNIVERSITY JOURNAL OF BIOLOGICAL SCIENCES (AUJBS)

Vol 1 No 1 March 2017 ISSN:

Reach at

VI Floor Science Block
Alagappa University
Karaikudi 630 003 Tamil Nadu India
alubiojournal@gmail.com

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Anti-bacterial activity, anti-inflammatory and anti- arthritic studies on mangroves by using in vitro model systems

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Received: 15.01.2017 / Accepted: 23.02.2017
Published online: 25.03.2017

Abstract The aim of the study was to investigate the antibacterial, anti-inflammatory and anti-arthritic activity of five mangrove plant extracts obtained from different solvent systems. Antibacterial activity of selected mangroves against human pathogens were screened by well diffusion technique. The extracts exhibited different degree of growth inhibition against tested bacterial strains. The methanol extract of *Avicennia marina* exhibited the highest anti-bacterial activity while chloroform extract did not show any growth inhibition. Phytochemical screening revealed that *A. marina* contained phenolic flavanoids, alkaloids, steroids, glycosides, tannins, saponnins and phlobatannins. In addition, the anti-inflammatory activities of different extracts of mangroves were investigated by human red blood cell membrane stabilization (HRBC) method. The methanol extract of *Avicennia marina* showed significant membrane stabilizing action on human red blood cell membrane compared with other mangroves. The HRBC membrane stabilization activity of the methanol extract of *A. marina* at concentration of 250µg/ml showed 81.24% inhibition of denaturation while standard diclofenac 100 µg/ml showed 98.65% inhibition of denaturation. In arthritic assay, methanolic extract of *A. marina* showed a greater percentage of inhibition of bovine serum albumin in denaturation at 89.64%. Thus the results indicate that methanolic extract of *A. marina* was found to possess significantly good antibacterial, anti-inflammatory and anti-arthritic activities and this can be attributed to the presence of phytochemical compounds. Further, *in vivo* studies are to be carried out to confirm their activity and explore the mechanism by which this plant acts in protecting from autoimmune disease and rationalize their use.

Key Words Growth inhibition; antibacterial activity; infectious pathogens; membrane stabilization assay; protein denaturation; bovine serum albumin

Introduction

Mangroves are biochemically unique, producing a wide array of novel natural products. Many mangrove plants have been used folklore medicine worldwide. Recently, extracts from mangroves and mangrove dependent species have been proven to possess antimicrobial activities against human, animal and plant pathogens. In spite of that, the traditional and ethno medicinal uses of these plants remain unexplored. Recent studies revealed the potential application of the mangroves in the discovery of novel antibiotics. However, a great deal of effort is required for compilation and dissemination of knowledge in this field. Such information may be valuable for the pharmaceutical industries for discovery of new therapeutic agents and antimicrobial drugs which can act effectively and efficiently against the broad spectrum of microorganisms.

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Inflammation is a natural defense mechanism of the human body against various external stimuli, injuries and infections however, it's a protective phenomenon, but when inflammatory conditions exceeds beyond its usual limit it can cause harm to the body and this term is usually called chronic Inflammation, which is the result of repeated prolonged infections and autoimmunity. Chronic inflammation is silent and persistent biological condition induced other inflammatory diseases like rheumatoid arthritis, atherosclerosis, cardiovascular diseases, diabetes and asthma, etc. Mangrove plant grows in the wetland of tropical and subtropical regions have a potential effect in many diseases like diabetes, inflammation, rheumatism, cardio vascular diseases etc.

Many modern drugs were originally extracted from plant sources, even if they are now made synthetically, and many other drugs are descended from plant substances. The inflammatory response involves a complex array of enzyme activation, mediator release, fluid extravasations, cell migration, tissue breakdown and repair which are aimed at host defense and usually activated in most disease conditions (Vane *et al.*, 1995). Currently, much interest has been paid in the searching of medicinal plants with anti-inflammatory activity which may lead to the discovery of new therapeutic agent that is not only used to suppress the inflammation but also used in diverse disease conditions where the inflammation response in amplifying the disease process. In this regard, we planned to work the various extracts of mangrove leaves were studied for its *in vitro* anti-bacterial activity, anti-inflammatory activities and anti-arthritic activity.

Materials and methods

Collection of mangrove plants

The mangrove plants, namely *Avicennia marina, Ceriops decandra, Exoecaria agallocha, Rhizophora mucronata, Soueta monoica* and *Soueta maritima*were collected from Karangadu mangrove forests (Latitude 9°38'58"N and Longitude 78°57'38"E). The leaves of the collected plants were cleaned with tap water for removing adhering sands and shade dried for 72 hrs under appropriate condition.

Extraction of bioactive compounds from collecting mangroves

Leaves of choosing plants were dried under shade condition. After drying, leaves of each plant were pulverized into fine powder using sterilized pestle and mortar. 500g of powdered leaves were soaked in 500ml of solvents; acetone, benzene, chloroform, ethanol and methanol for 10 days in an air tight clean glass container. This method is called percolation. Mixing was done by everyday that enhance maximum bioactive compounds from mangroves. After that, the solvent with bioactive compounds were filtered with whatman filter paper. The filtrate was kept on a rotary flash evaporator and concentrated to dryness at 45ºC. The extracts were stored at 4ºC for further screening. The suspended materials were dried under room temperature and weighed. The percentage of extraction was calculated as follows:

$$\%\ \text{of extraction} = \frac{\text{Weight of the extract (g)}}{\text{Weight of the plant material}}$$

Antibacterial activity of choosing mangrove extracts against human pathogens

Extracts of *Avicennia marina, Ceriops decandra, Exoecaria agallocha, Rhizophora mucronata, Soueta monoica* and *Soueta maritima*were tested against human pathogens

namely *Streptococcus*, *Staphylococcus*, *Solmonella*sp. *Shigella*, *Vibrio* and *E. coli* by welldiffusion assay. The antibacterial assay was carried out by the agar well diffusion method using the Muller Hinton agar medium (Himedia Pvt Ltd Mumbai). The microorganisms were activated by inoculating a loopful of the strain in nutrient broth (20 ml) incubated at 37ºC for 24 hrs. Then 0.1 ml of the fresh inoculums was swabbed uniformly over the sterile Muller Hinton agar by using a sterile cotton swab. Three Well (6mm diameter) are made on the seeded plates with the sterilized cork borer. In each well, different solvent extracts of choosing mangrove plants were loaded by using micropipette at concentration of 250 mg/ml. Separately, six plates with sterile medium placed bacterial strains and walls made are loaded with pure solvent served as control. Later plates were placed at room temperature for an hour to allow diffusion of extract into the agar and incubated at 37ºC for 24 hrs. All the tests were carried out in triplicate and the average values for zones of inhibition in millimeters (mm) were taken over 24 hours of incubation.

***In vitro* anti-inflammatory activity of choosing mangrove plants**

The Human red blood cell (HRBC) membrane stabilizing method

The anti-inflammatory activity of various extracts of choosing mangroves was assessed by HRBC membrane stabilization method. Blood was collected from healthy volunteers. The collected blood was mixed with equal volume of Alsever solution (dextrose 2%, sodium citrate 0.8%, citric acid 0.05%, sodium chloride 0.42%, and distilled water 100 ml) and centrifuged with isosaline. To 1ml of HRBC suspension, equal volume of test drug in five different concentrations 50, 100, 150, 200 and 250 g/ml were added. All the assay mixtures were incubated at 37ºC for 30 minutes and centrifuged. The haemoglobin content in the supernatant solution was estimated by using spectrophotometer at 560 nm (Sandhya*et al.*, 2010). The percentage of haemolysis was calculated then by the formula as given below:

$$\text{Percent of hemolysis} = \frac{\text{OD of test sample}}{\text{OD of control}} \text{ X } 100$$

Here "OD of test" is optical density or the test sample absorbance and "OD of control" is optical density or observance of the negative control.

In vitro anti arthritic activity of chosen mangrove plant

Inhibition of protein denaturation assay

Prepared the reaction mixture (0.5 ml) consisted of 0.45 ml bovineserum albumin (5% aqueous solution) and 0.05 ml of chosen mangrove extracts to observe the protein denaturation in the sample. pH was adjusted at 6.3 using a small amount of 1N HCl. The samples were incubated at 37°C for 20 min and then heated at 57°C for 30 min. After cooling the samples, 2.5 ml phosphate buffer saline (pH 6.3) was added to each tube. Turbidity was measured spectro photometrically at 660. Control test, 0.05 ml distilled water was used instead of extracts while product control test lacked bovine serumalbumin. The percentage inhibition of protein denaturation was calculated as follows.

Percent inhibition = 100 - (O.D. of test – O.D. of product control) X 100

Qualitative analysis of phytochemical constituents of chosen mangrove plants

The different qualitative chemical tests were carried out using the standard procedures

to identify the constituents as described by Edeoga *et al.*, (2005) as follows:

Phytochemcial Tests	Procedure
Test for Phenolic Flavonoids	One ml of filtrate with 2 ml of 10% lead acetate gives brown precipitate.
Test for Alkaloids	One ml of the filtrate with 2 ml of Drangendroff's reagent shows turbid orange colour.
Test for Terpenoids	One ml of the filtrate with 2 ml *CHC* is carefully added to few drops of concentrated . An interface with a reddish brown coloration is formed.
Test for Steriods	To one ml of the filtrate, 10 ml of the chloroform and 10 ml of is slowly added by the side s of the test tube. Upper layer turns red and sulphuric acid layer showed yellow with green fluroscences.
Test for Cardiac Glycosides	To one ml of the filtrate, one ml of FeC regeant is added (mixture of 1 volume 0.5% FeC solution + 99 volume of glacial acetic acid) with few drops of concentrated . Greenish blue color appears within few minutes.
Test for Tannis	One ml of filtrate with 2 ml of ferric chloride gives dark green color.
Test for Flavonoids	One ml of filtrate with of dilute NaOH Shows development of golden yellow color.
Test for Saponins	One ml of filtrate with 2 ml distilled water, is shaken vigorously and allowed to stand for 10 minutes. Development of foam on the surface of mixture, lasting for 10 minutes indicates the presence of saponins.
Test for Anthraquinones	To one ml of the filtrate, 10 ml benzene was added, followed by filtering and addition of 5ml of 10%(v/v) ammonia to the filtrate and shake well. Development of pinkish colored solution indicates the presence of anthraquinones.
Test for Anthocyanides	Add one ml of filtrate with 5ml of dilute HCl, the appearance of pale pink color indicates the presence of the above compound.
Test for Phlobatanins	To one ml of the filtrate, few drops of 1% aqueous HCl is added. A red precipitate is formed.

Results and discussion

One of the major challenges in drug development is the choice of extracts that should be used to isolate the compound(s) and the choice of solvent depends on what intended with the extract. In the present study, five different solvents namely acetone, benzene, chloroform, ethanol and methanol were used for preparation of extracts from selected mangroves. The present results revealed that methonol was relatively the best solvent for extraction of leaves principle constitutes compared to the extracts obtained with other solvents. Methanolic extract of *Avicennia marina* was reported high yield of extracts compare with other mangroves in different solvents (Table 1).

The results of the phytochemical analysis showed that phenolic flavanoids and alkaloids are present in the methanolic extracts of chosen mangroves (Table 2). These phytochemicals have a wide variety of antimicrobial properties against pathogenic fungal and bacterial strains (Abeysinghe *et al.*, 2002). According to literature, alkaloids, triterpenoides, flavonoids, saponins are the major classes of antioxidants in plants (Saad *et al.*, 2011). In the present study, phytochemical analyses showed that extracts of chosen

mangroves contain most of principles are an indicative for the antibacterial activity and antiarthritic activities. Therefore, the beneficial medicinal effects of plant materials may results from the combination of antimicrobial, antioxidants and other secondary products present in the plant, as well as phytochemicals (Cai *et al.*, 2004). Such secondary products play an important role in a plant's defense through cytotoxicity towards microbial pathogens and this could prove the usefulness of these secondary products as antimicrobial medicines for humans (Tukiran, 2013).

Antibacterial activity

The results of antibacterial activity of the six mangrove plants were extracted in various solvents are given in table 3. The methanolic extract of mangroves inhibited the growth of all the tested pathogens than the other solvent extracts of chosen mangroves. Among the mangrove species *Avicennia marina* showed maximum antibacterial activity, with a mean zone of inhibition high when compared to the all pathogens to each mangrove. In contrast, *Soueta maritima* of the tested mangrove showed very less antibacterial activity compared with other mangroves. The antibacterial activity exhibited by the mangrove plant extracts could be due to the presence of phytochemicals such as alkaloids, tannins and flavonoids medicine to treat various diseases for centuries (Kathiresan, 2000). The present study indicates that when compared with chlorofrom extracts, the crude methanol extracts showed better inhibition activity against all tested pathogens indicating that active ingredients in the plant could be extracted in to methanol. However some plants extracts did not exhibit any activity. This is probably because accumulation of intracellular active ingredients is low or the concentration of the compound used may not be sufficient. Some of the mangrove plants have been screened earlier for antiviral, antibacterial, antiulcer and anti-inflammatory activity (Premnathan *et al.*, 1992). The plant is being tested for modern medical uses and clinical trials show that the plant may have anti-HIV, anti-cancer, anti-bacterial and anti-viral properties. Hence in recent years screening of mangrove plants is being paid more attention to develop novel drugs against pathogens.

Anti-inflammatory studies

There are certain problem in using animals in experimental pharmacological research such as ethical issues and lack of rationale for their use when other suitable methods are available or could be investigated. Hence the present study was selected *in vitro* model system to screen the anti-inflammatory and anti-arthritic property from chosen mangroves. In anti-inflammatory activity, stabilization of RBCs membrane was studied to further develop the mechanism of anti inflammatory action of methanolic extract of chosen mangroves. These extracts were effectively inhibited the heat induced haemolysis at different concentration. These provide evidence for membrane stabilization as an additional mechanism of their anti-inflammatory effect. The selected mangroves extracts may possible inhibit the release of lysosomal content of neutrohils at the site of inflammation. These neutrophil lysosomal constituents include bactericidal enzymes and protease, which upon extracellular release cause further tissue inflammation and damage (Chou, 1997). The percentage of membrane stabilization for methanolic extracts and diclofenac were done at 50, 100, 150, 200 and 250µg/ml. The present results shows that, methanolic extract of *Avicennia marina* are

effective in maximum inhibiting the heat induced hemolysis of HRBC at 250 µg/ml with the increasing concentration the membrane hemolysis is decreased as represented in table 4. Hence anti-inflammatory activity of the extracts was concentration dependent. In addition diclofenac showed the maximum inhibition (98.65%) at 100 µg/ml. *Avicennia marina* has significant anti-inflammatory activity which may be due to presence of chemical profile such as phenolic flavanoids and alkaloids.

Although the precise mechanism of this membrane stabilization is yet to be elucidated, it is possible that *Avicennia marina* produced this effect surface/volume ratio of the cells, which could be brought about by an expansion of membrane or the shrinkage of the cells and an interaction with membrane proteins (Shinde *et al.*, 1997).

Anti-arthritic assay

In vitro anti arthritic assay was performed using the most popular method as inhibition of protein denaturation assay with different concentration of mangrove extracts were tested to find out the percentage inhibition. The results were found that the methanolic extract of *Avicennia marina* at concentration of 250µg/ml showed maximum percentage inhibition (89.63%). A dose dependent increase in the percentage inhibition was observed for all the concentration tested with a dose dependent increase in anti-arthritic activity (Table 5).In this method diclofenac was also used as standard for comparing its anti-arthritic activity and when compared to mangrove extract. It was showed 92.65% inhibition at concentration of 100 µg/ml.

In protein denaturation assay, the denaturation of BSA is induced by heat treatment. The denatured BSA express antigen associated to type III hypersensitivity reaction which are related to diseases such as serum sickness, glomerulo nephritis etc.(Fayyaz Ahmad *et al.*, 1992). Heat denatured proteins are as effective as native proteins in provoking delayed hypersensitivity (Gell and Benacerraf, 1959). Moreover, it was already proved that conventional NSAID's like phynylbutazone and indomethazine does not act only by the inhibition of endogenous prostaglandins production by blocking Cox enzyme but also by prevention of denaturaion of proteins (Insel, 1990). Thus protein denaturation assay is the convenient method to check the anti arthritic activity. Denaturation of proteins is a well documented cause of inflammation and rheumatoid arthritis (Mizushima, 1966). Several anti-inflammatory drugs have shown dose dependent ability to inhibit thermally induced protein denaturation (Grant, 1970). Ability of *Canthium parviflorum* extract to bring down thermal denaturation of protein is possibly a contributing factor for its anti-inflammatory activity. The anti-inflammatory activity of *Canthium parviflorum* extract found may be due to the presence of therapeutically active flavonoids. The therapeutic applications of flavonoids on inflammation have previously been reported (Middleton, 2000; Havsteen, 2002). The data of our studies suggests that selected mangroves showed significant anti-inflammatory activity by *in vitro* methods tested. Further studies involving the isolation and purification of the chemical constituents of the plant and the investigations in the biochemical pathways may result in the development of a potent antibacterial, anti-inflammatory and anti-arthritic agent without toxicity and better therapeutic index.

Conclusion

Mangroves have been valued for centuries in ayurvedic medicine. Phyto-chemical analysis of mangrove plant extracts revealed the presence of various bio-chemical compounds such as flavonoids, glycosides, alkaloids, saponins and terpenoids. Since glycosides and flavonoids have remarkable anti-inflammatory activity, our present work aims at evaluating the antibacterial, *in-vitro* anti-inflammatory activity and anti- arthritic activity of mangroves by *in vitro* model systems. The data of our studies suggests that methanolic extracts of mangroves showed significant activity to all screening methods. Therefore our studies support the isolation and use of active constituents of mangroves in treating bacterial infection and inflammation.

Acknowledgements

The authors wish to thank The Secretary and The Principal of Dr. Zakir Husain College, Ilayangudi – 630 702, Sivagangai District, Tamil Nadu, India for providing necessary facilities to carry out this study.

Table 1: Yield of crude extracts on mangroves from different solvent systems

S. No.	Name of the mangroves	Weight of the dry sample of mangrove plants (g)	Yield of crude extracts (%; w/w)				
			A	B	C	E	M
1.	*Avicennia marina*	500	13	20	29	26	32
2.	*Ceriops decandra*	500	15	18	09	11	22
3.	*Exoecaria agallocha*	500	08	13	16	09	11
4.	*Rhizophora mucronata*	500	06	13	16	12	16
5.	*Soueta monoica*	500	10	14	14	12	24
6.	*Soueta maritima*	500	12	07	13	10	13

A-Acetone; B-Benzene; C-Chloroform; E- Ethanol; M- Methanol

Table 2: Phytochemical analysis of chosen mangrove plant extracts obtained from methanol

S.No.	Phytochemicals	Observation					
		A. marina	*C. decandra*	*E. agallocaha*	*R. mucronata*	*S. monoica*	*S. Maritima*
1.	Phenolic Flavonoids	+	+	+	+	+	+
2.	Alkaloids	+	+	+	+	+	+
3.	Terpenoids	-	-	-	-	+	-
4.	Steroids	+	+	-	-	+	+
5.	Glycosides	+	-	+	+	-	-
6.	Tannis	+	-	-	+	+	+
7.	Flavonoids	-	+	+	-	-	-
8.	Saponins	+	+	+	-	-	+
9.	Anthraquinones	-	-	-	-	-	-
10.	Anthocyanides	-	-	-	-	-	-
11.	Phlobatannis	+	+	+	+	-	-

+ Presence; - Absence

Table 3: Antibacterial activity of mangrove extracts in different solvent systems

Name of the mangroves	Solvent systems	Zone of inhibition in diameter (mm)					
		Streptococcus	*Staphylococcus*	*E. coli*	*Salmonella*	*Vibrio*	*Shigella*
Avicennia marina	Acetone	-	12	3	12	-	-
	Benzene	-	-	-	-	-	-
	Chloroform	-	-	-	-	-	-
	Ethanol	16	-	9	8	-	10
	Methanol	18	15	16	14	09	12
Ceriops decandra	Acetone	-	-	-	-	-	-
	Benzene	-	-	-	-	-	-

	Chloroform	-	9	8	-	10	-
	Ethanol						
	Methanol	15	16	14	07	12	15
Exoecaria agallocha	Acetone	-	12	3	12	-	-
	Benzene	16	-	9	8	-	10
	Chloroform	11	15	16	14	-	12
	Ethanol	-	-	-	-	-	-
	Methanol	19	09	12	16	10	08
Rhizophora mucronata	Acetone	-	9	8	-	10	-
	Benzene						
	Chloroform	15	16	14	-	12	15
	Ethanol	-	12	3	12	-	-
	Methanol	16	13	9	8	10	10
Soueta monoica	Acetone	-	12	3	12	-	-
	Benzene						
	Chloroform	-	-	-	-	-	-
	Ethanol	-	-	-	-	-	-
	Methanol	11	12	10	14	08	12
Soueta maritima	Acetone	-	-	-	-	-	-
	Benzene	-	-	-	-	-	-
	Chloroform	-	-	-	-	-	-
	Ethanol	-	-	-	-	-	-
	Methanol	12	09	07	05	06	12

Table 4: Anti-inflammatory activity of chosen mangroves with different concentration by RBC membrane stabilization assay

Name of the mangroves	Concentration (µg/ml)	% of inhibition
Avicennia marina	50	52.31
	100	62.30
	150	75.98
	200	78.69
	250	81.24
Ceriops decandra	50	10.26
	100	32.65
	150	45.6
	200	62.15
	250	69.25
Exoecaria agallocha	50	23.65
	100	39.31
	150	52.36
	200	65.21
	250	70.12
Rhizophora mucronata	50	29.57
	100	36.15
	150	45.27
	200	59.34
	250	76.45
Soueta monoica	50	28.16
	100	45.61
	150	68.34
	200	75.31
	250	79.16
Soueta maritima	50	35.61
	100	42.19
	150	64.28

	200	74.12
	250	79.12
Dicofenac	100	98.65
Control	100	-

Table 5: *In vitro* anti – arthritic activity of mangroves with different concentrations by using protein denaturation assay

Name of the mangroves	Concentration (µg/ml)	% of inhibition
Avicennia marina	50	56.23
	100	68.16
	150	75.68
	200	85.16
	250	89.64
Ceriops decandra	50	26.34
	100	36.97
	150	46.28
	200	52.64
	250	69.78
Exoecaria agallocha	50	23.19
	100	36.31
	150	41.26
	200	62.34
	250	70.12
Rhizophora mucronata	50	32.16
	100	40.28
	150	61.27
	200	69.15
	250	71.28
Sueta monoica	50	20.16
	100	35.91
	150	49.67
	200	51.28
	250	68.15
Sueta maritima	50	10.45
	100	23.67
	150	46.15
	200	52.14
	250	59.17
Dicofenac	100	92.65
Control	100	-

References

Abeysinghe, P.D., Pathirana, R. N., and Wanigatunge, R. P., 2012. Evaluation of antibacterial activity of different mangrove plant extracts. Ruhuna journal of science 1, 1-5.

Ahmad, Fayyaz, Rafeeq, A. Khan, and Shahid Rasheed., 1992. Study of analgesic and anti-inflammatory activity from plant extracts of Lactuca scariola and Artemisia absinthium. Journal of Islamic Academy of Sciences 52, 111-14.

Benacerraf, B., and Gell, P. G. H., 1959. Studies on Hypersensitivity - I. delayed and drthus-type skin reactivity to protein conjugates in Guinea Pigs." Immunology 21, 53.

Cai, Yizhong, , 2004. Antioxidant activity and phenolic compounds of 112 traditional Chinese medicinal plants associated with anticancer. Life sciences 74 (17), 2157-2184.

Chou, Chung-Tei., 1997. The Anti-inflammatory effect of an extract of Tripterygium wilfordii Hook F on Adjuvant-induced Paw Oedema in rats and inflammatory mediators release." Phytotherapy Research 11(2), 152-154.

Edeoga, H. O., Okwu, D. E., and Mbaebie, B. O., 2005. Phytochemical constituents of some Nigerian medicinal plants. African journal of biotechnology 4(7), 685-688.

Grant, Norman, H., Harvey, E. Alburn, and Catherine Kryzanauskas., 1970. Stabilization of serum albumin by anti-inflammatory drugs. Biochemical Pharmacology 19 (3), 715-722.

Havsteen, Bent, H., 2002. The biochemistry and medical significance of the flavonoids." Pharmacology & therapeutics 96(2), 67-202.

Insel, T. R., Miller, L. P. and Gelhard, R. E., 1990. The ontogeny of excitatory amino acid receptors in rat forebrain—IN-methyl-d-aspartate and quisqualate receptors. Neuroscience 35(1), 31-43.

Kathiresan, K., 2000. A review of studies on Pichavaram mangrove, southeast India.Hydrobiologia 430(1-3), 185-205.

Middleton, Elliott, Chithan Kandaswami, and Theoharis Theoharides., C., 2000. The effects of plant flavonoids on mammalian cells: implications for inflammation, heart disease, and cancer.Pharmacological reviews 52(4), 673-751.

Mizushima, Y., and M. Kobayashi., 1968. Interaction of anti-inflammatory drugs with serum proteins, especially with some biologically active proteins. Journal of Pharmacy and Pharmacology 20(3), 169-173.

Premnathan, M., 1992. A survey of some Indian marine plants for antiviral activity. Botanica marina 35(4), 321-324.

Saad, Shahbudin, et al., 2011. Antimicrobial activity of mangrove plant (Lumnitzera littorea)." Asian Pacific

Journal of tropical medicine 4(7), 523-525.

Sandhya, S., 2010. Membrane stabilizing potency of two Tephrosia species." Journal of Phytology 2(6), 1-5.

Shinde, U. P., J. J. Liu, and M. Inouye., 1997. Protein memory through altered folding mediated by intramolecular chaperones. Nature 389(6650), 520-522.

Tukiran, B., 2013. Phytochemical analysis of some plants in Indonesia. J. Biol. Agric. Healthcare,3, 6-10.

Vane, J. R., and Botting, R. M., 1995. New insights into the mode of action of anti-inflammatory drugs. Inflammation Research 44(1), 1-10.